Exploring the RCP Database

Group: _____ Names: _____

> Database: https://tntcat.iiasa.ac.at/RcpDb/. Each group completes its own section and posts its answers on the class Miro board. All groups then answer the closing question.

Group A — Concentrations (Region: World)

Hint: under Variable, look for Concentration, then the gas.

Gas	2005	RCP2.6 (2100)	RCP4.5 (2100)	RCP6.0 (2100)	RCP8.5 (2100)
CO_2 (ppm)					
CH_4 (ppb)					
N_2O (ppb)					

Aa. For CO_2 and CH_4, compare the 2100 value in each pathway with the 2005 value. Which gas falls in any pathway, and why can it fall when the other cannot?

Ab. Which gas changes least across the four pathways? Where do its emissions come from?

Group B — Greenhouse gas emissions (Region: World)

Hint: Variable → Emissions → CO_2 (fossil and industrial). The units are GtC per year, not $GtCO_2$, so multiply by 3.67 to convert.

CO_2 fossil and industrial (GtC/yr)	2010	Peak year	2100
RCP2.6			
RCP4.5			
RCP6.0			
RCP8.5			

Ba. Which pathway turns net negative, and in roughly which year? What processes remove the CO_2?

Bb. Compare CH_4 and N_2O emissions across the four pathways. Which changes least, and why?

Group C — Air pollutants (Region: World)

Hint: Variable → Emissions → Sulfur (SO_2), BC (black carbon) or NOx. The units are Tg per year.

Pollutant	2000 to 2010 level	2100, by RCP
SO_2		
Black carbon		
NOx		

Ca. How do SO_2 emissions change by 2100 in each RCP, including RCP8.5? What does that change imply for aerosol cooling?

Cb. Do black carbon and NOx follow the same path as CO_2? Could a city such as Lahore reduce its smog while still emitting large amounts of CO_2?

Group D — Regions (emissions only)

Hint: change Region to the region codes below and choose Variable → Emissions → CO_2 (fossil and industrial). OECD90: the old OECD countries; ASIA: developing Asia, including Pakistan; MAF: Middle East and Africa; REF: Former Soviet Union and Eastern Europe; LAM: Latin America.

CO_2 (GtC/yr)	2010	RCP2.6 (2100)	RCP8.5 (2100)
OECD90			
ASIA			
MAF			
REF			

Da. Compare OECD90 and ASIA under both pathways. Calculate ASIA ÷ World for 2010 and for 2100. Does Asia's share rise or fall?

Db. Which region grows fastest under RCP8.5? Pakistan is not listed separately: what data would be needed to estimate its share?

Group E — Forcing and models (Region: World)

Hint: Variable → Forcing → Total radiative forcing (W/m^2). For the model names, look at the scenario names or the documentation pages.

Forcing (W/m^2)	2010	2050	2100	Model and team
RCP2.6				
RCP4.5				
RCP6.0				
RCP8.5				

Ea. When does RCP2.6 forcing peak, and at what level? What happens to emissions at the same time?

Eb. Each RCP was produced by a different model. Can the four RCPs be read as one world under four different policies? Explain.

Closing question (all groups)

Suppose your group is advising a planner in Pakistan. What information would the planner need that the RCP Database does not provide?